AIML LABORATORY — MACHINE LEARNING MINI PROJECT

ActivitySense AI

Smartphone Sensor-Based Human Activity Recognition using the UCI HAR Dataset

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Research questions: (RQ1) Which of KNN, SVM and Random Forest best balances classification performance and computational cost? (RQ2) How informative are accelerometer vs gyroscope features? (RQ3) Can a reduced feature subset match the full 561 features? (RQ4) Which activities are most often confused?

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1. Aim

To build and compare machine-learning models that recognise six human activities (walking, walking upstairs, walking downstairs, sitting, standing, laying) from smartphone accelerometer and gyroscope features, and to serve the trained models through an interactive application.

2. Problem statement

Smartphones contain accelerometers and gyroscopes whose signals differ between activities. This project uses supervised machine learning to recognise activities from sensor-derived features and lets a user obtain a predicted activity with model-estimated confidence scores.

3. Dataset

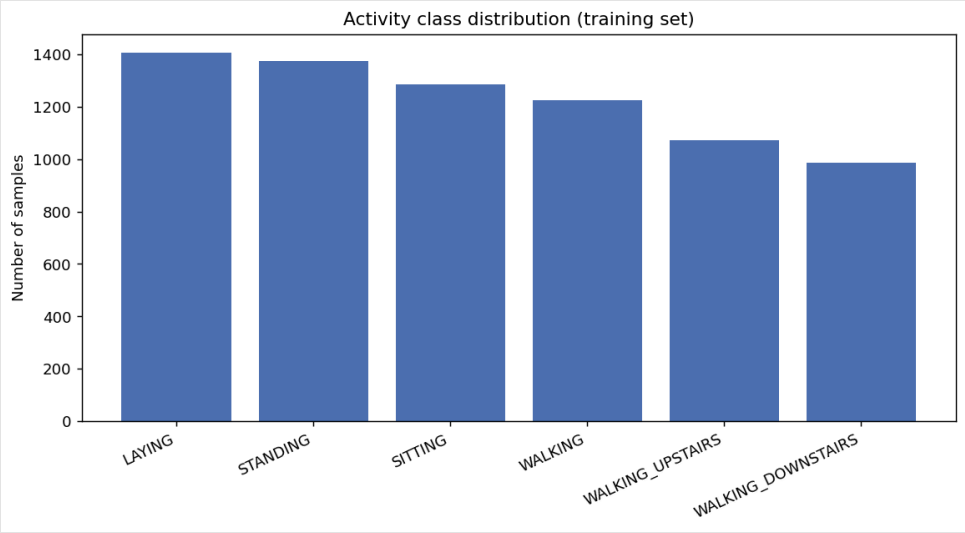
The **UCI HAR** dataset: 30 volunteers wearing a waist smartphone, signals sampled at 50 Hz and processed into fixed windows, yielding **561 engineered features** and six activity classes. It provides an official **subject-wise** split of **7352 training** and **2947 test** samples, which we preserve.

4. Methodology

- Loaded the official files; assigned 561 feature names (84 duplicate names made unique).
- Validated the data (no missing/non-finite values; train/test columns match).
- **Leakage-safe pipelines:** `StandardScaler` inside each KNN/SVM pipeline (fit on training folds only); Random Forest needs no scaling.
- **Four feature groups:** full (561), accelerometer (350), gyroscope (213), and a reduced subset of 50 chosen with `SelectKBest` (mutual information) on training data.
- Tuned each model with `GridSearchCV` on the training partition; evaluated once on the official test set. Recorded accuracy, macro-/weighted-F1 and train/predict time.

5. Exploratory Data Analysis

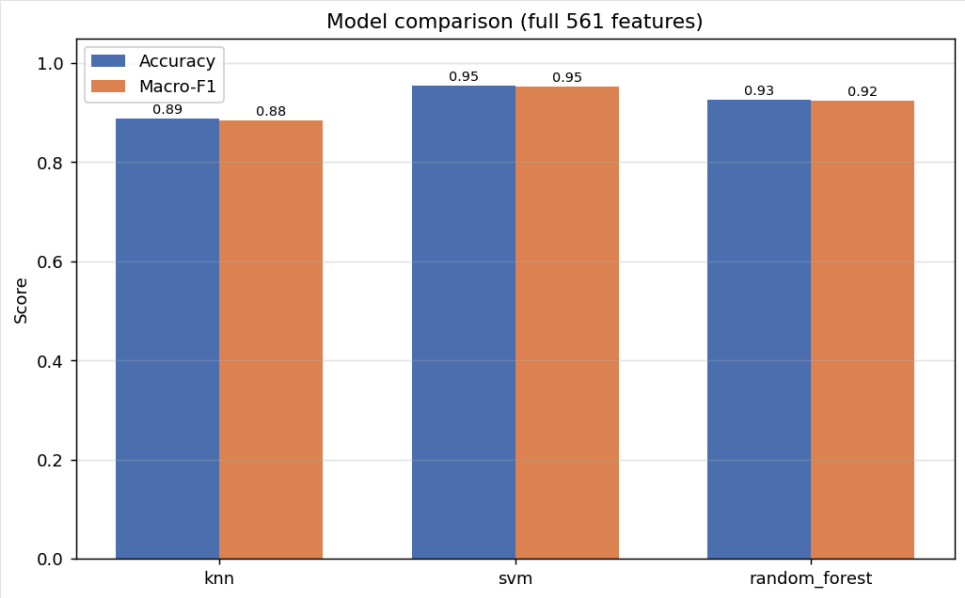
The six activities are fairly balanced and split into dynamic (walking types) and static (sitting/standing/laying):



6. Results

6.1 Model comparison (RQ1) — full 561 features

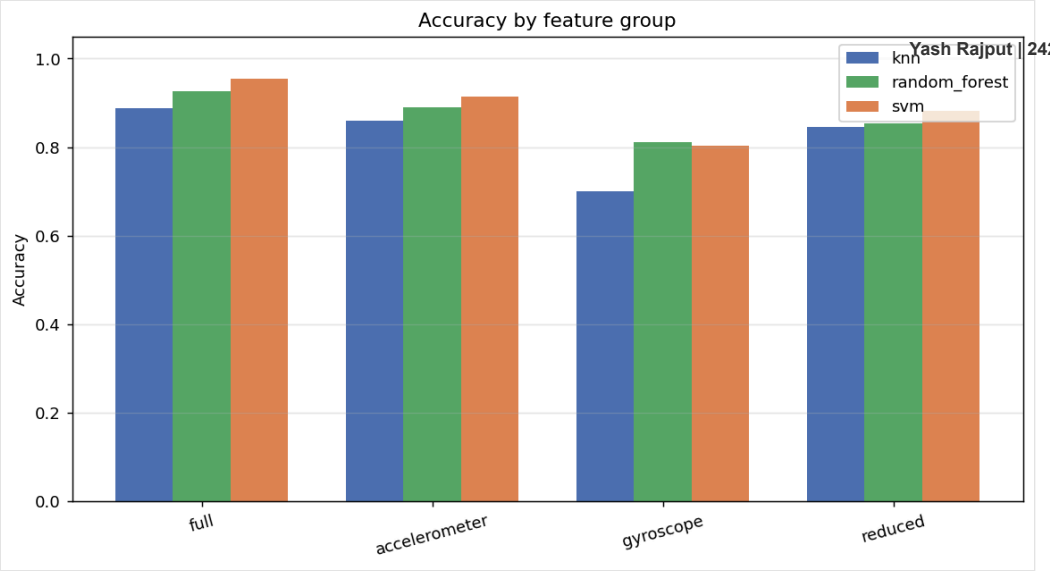
model	accuracy	macro_f1	weighted_f1	train_time_s	predict_time_s
knn	0.888	0.885	0.887	0.260	2.948
svm	0.954	0.953	0.954	5.553	4.983
random_forest	0.926	0.924	0.926	16.756	0.146



SVM is the most accurate (95.4%); **Random Forest** is close and predicts the fastest; **KNN** is weakest and slow at inference (it compares each test point to all training points).

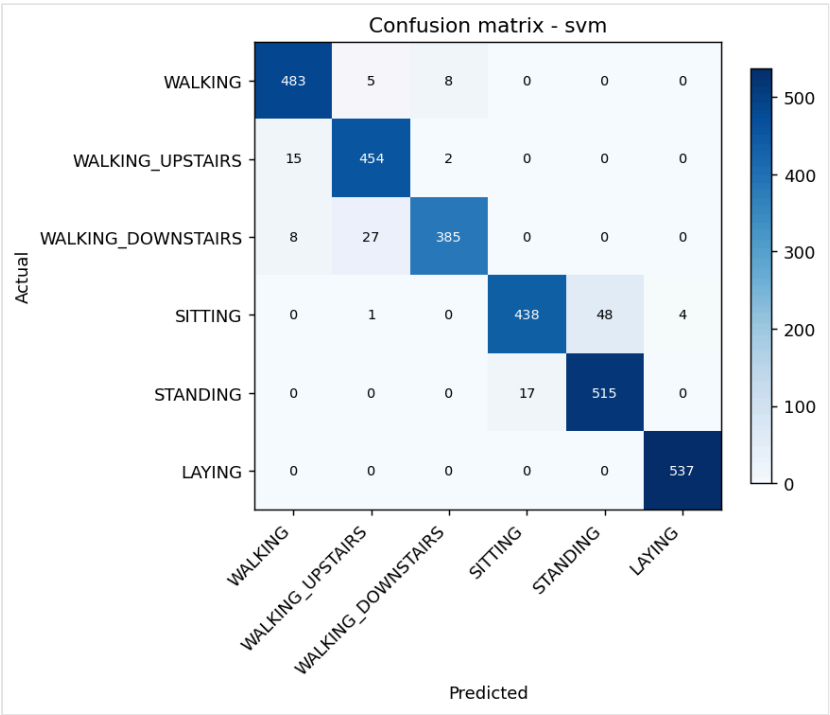
6.2 Feature groups (RQ2 & RQ3)

model	n_features	knn	random_forest	svm
feature_group				
full	561	0.888	0.926	0.954
accelerometer	350	0.859	0.890	0.914
gyroscope	213	0.700	0.811	0.803
reduced	50	0.845	0.854	0.881



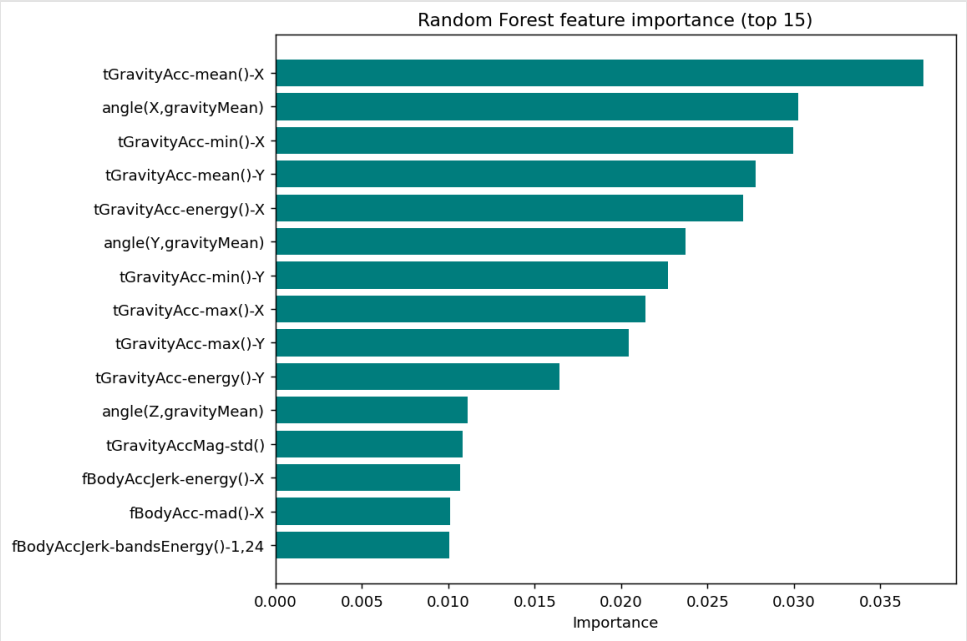
RQ2: accelerometer features alone are far more informative than gyroscope features alone for every model. **RQ3:** a reduced 50-feature subset retains most of the full-feature accuracy (SVM 0.881 vs 0.954), i.e. a much smaller model is nearly as good.

6.3 Activity confusion (RQ4)



Errors concentrate among the **static** activities — mainly **sitting vs standing** — while the dynamic/static boundary is almost never crossed.

6.4 Feature importance



7. Interactive application

A Streamlit app serves the trained models over five pages (Home, Predict Activity, Model Comparison, Feature Analysis, Prediction History). Prediction supports three inputs: selecting a held-out test example (shows actual vs predicted), uploading a 561-feature CSV (validated, with downloadable results), and manual entry using a separate small-feature model (a demonstration). Scores are labelled as model-estimated, not calibrated probabilities.

8. Conclusion

Three classifiers were compared on the UCI HAR benchmark with the official subject-wise split and leakage-safe pipelines. SVM gave the best accuracy (95.4%), Random Forest traded a little accuracy for the fastest inference, and KNN was weakest. Accelerometer features proved far more informative than gyroscope features, and a 50-feature subset retained most of the accuracy of the full 561. The main remaining confusion is between sitting and standing. Because the UCI features are already heavily engineered, even simple models perform well; the project adds value through the model/feature comparison and the interactive prediction app.